

Task 1: Write a C program to implement Producer–Consumer communication using an anonymous pipe. Transfer 10,000 messages from the producer to the consumer, calculate the total bytes transferred, measure the execution time, and display the communication rate in messages per second.

```
karthik@Karthikeya:~$ mkdir -p ~/producer_consumer_pipe
karthik@Karthikeya:~$ cd ~/producer_consumer_pipe
karthik@Karthikeya:~/producer_consumer_pipe$ nano pipe.c
karthik@Karthikeya:~/producer_consumer_pipe$ gcc pipe.c -o pipe
karthik@Karthikeya:~/producer_consumer_pipe$ ./pipe
Child Consumer:
Messages consumed : 100000
Total bytes       : 3888895

=====
Producer-Consumer Pipe Communication
=====
Messages transferred : 100000
Time taken           : 0.054414 seconds
Communication rate   : 1837762.34 messages/sec
=====
karthik@Karthikeya:~/producer_consumer_pipe$ |
```

Task 2: Write a C program to implement a pipeline using ls and grep to filter and display matching filenames. Compile and execute the program in Ubuntu and verify the output.

```
karthik@Karthikeya:~/producer_consumer_pipe$ mkdir -p ~/ls_grep_pipe
karthik@Karthikeya:~/producer_consumer_pipe$ cd ~/ls_grep_pipe
karthik@Karthikeya:~/ls_grep_pipe$ nano ls_grep.c
karthik@Karthikeya:~/ls_grep_pipe$ gcc ls_grep.c -o ls_grep
karthik@Karthikeya:~/ls_grep_pipe$ ./ls_grep
-rw-r--r-- 1 karthik karthik 1500 Sep 20 15:15 ls_grep.c
karthik@Karthikeya:~/ls_grep_pipe$ touch test1.c test2.c hello.c
karthik@Karthikeya:~/ls_grep_pipe$ ./ls_grep
-rw-r--r-- 1 karthik karthik 0 Sep 20 15:16 hello.c
-rw-r--r-- 1 karthik karthik 1500 Sep 20 15:15 ls_grep.c
-rw-r--r-- 1 karthik karthik 0 Sep 20 15:16 test1.c
-rw-r--r-- 1 karthik karthik 0 Sep 20 15:16 test2.c
karthik@Karthikeya:~/ls_grep_pipe$ |
```